

# PROJECT REPORT

## Design and Implementation of a Web Server on Raspberry Pi

### Description

This project demonstrates the development of a web server using the Raspberry Pi. The Raspberry Pi acts as a mini computer that hosts a web application, allowing users to access and control connected devices through a browser over a local network. It shows how embedded systems can be used to create low-cost and efficient web-based control systems.

### Components Used

Raspberry Pi 4  
Micro SD Card (16GB or higher)  
Power Adapter  
Ethernet Cable / WiFi Connection

### Program

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Alex Morgan Portfolio</title>

<!-- Google Font -->

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;700&display=swap"
rel="stylesheet">

<style>

*{

margin:0;

padding:0;
```

```
    box-sizing:border-box;

    font-family: 'Poppins', sans-serif;
}
```

```
body{

    background:#f5f5f5;

    color:#333;
}
```

```
/* Navbar */

header{

    background:#0b1c2d;

    padding:20px 10%;

    display:flex;

    justify-content:space-between;

    align-items:center;
}
```

```
.logo{

    color:#fff;

    font-size:28px;

    font-weight:600;
}
```

```
nav a{

    color:#fff;

    text-decoration:none;

    margin-left:30px;

    font-weight:400;

    transition:0.3s;
}
```

```
nav a:hover{  
    color:#4f8cff;  
}
```

```
/* Hero Section */
```

```
.hero{  
    height:90vh;  
    background: linear-gradient(135deg,#4f8cff,#6a5cff);  
    display:flex;  
    flex-direction:column;  
    justify-content:center;  
    align-items:center;  
    text-align:center;  
    color:#fff;  
    padding:0 20px;  
}
```

```
.hero h1{  
    font-size:55px;  
    font-weight:700;  
    margin-bottom:20px;  
}
```

```
.hero p{  
    font-size:20px;  
    margin-bottom:30px;  
}
```

```
.btn{  
    padding:15px 35px;
```

```
background:#111;
color:#fff;
border:none;
border-radius:40px;
font-size:16px;
cursor:pointer;
transition:0.3s;
}
```

```
.btn:hover{
    background:#333;
}
```

```
/* Section General */
section{
    padding:80px 10%;
    text-align:center;
}
```

```
section h2{
    font-size:36px;
    margin-bottom:30px;
}
```

```
/* About */
.about p{
    max-width:700px;
    margin:auto;
    font-size:18px;
    line-height:1.8;
}
```

```
/* Services */
```

```
.services-container{  
    display:flex;  
    justify-content:space-between;  
    gap:20px;  
    margin-top:40px;  
    flex-wrap:wrap;  
}
```

```
.service-box{  
    background:#fff;  
    padding:40px 20px;  
    border-radius:15px;  
    flex:1;  
    min-width:280px;  
    box-shadow:0 5px 15px rgba(0,0,0,0.05);  
    transition:0.3s;  
}
```

```
.service-box:hover{  
    transform:translateY(-10px);  
}
```

```
.service-box h3{  
    margin-bottom:15px;  
}
```

```
/* Portfolio */
```

```
.portfolio{  
    background:#f0f0f0;
```

```
}
```

```
/* Responsive */
```

```
@media(max-width:768px){
```

```
  .hero h1{
```

```
    font-size:35px;
```

```
  }
```

```
  nav{
```

```
    display:none;
```

```
  }
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
  <div class="logo">Alex Morgan</div>
```

```
  <nav>
```

```
    <a href="#">Home</a>
```

```
    <a href="#about">About</a>
```

```
    <a href="#services">Services</a>
```

```
    <a href="#portfolio">Work</a>
```

```
    <a href="#">Contact</a>
```

```
  </nav>
```

```
</header>
```

```
<section class="hero">
```

```
  <h1>UI Designer Crafting Beautiful Digital Experiences</h1>
```

```
  <p>I design modern, user-centered interfaces for web and mobile applications.</p>
```

```
  <button class="btn">View My Work</button>
```

</section>

<section class="about" id="about">

<h2>About Me</h2>

<p>

I'm a passionate UI Designer with 5+ years of experience creating visually appealing and user-friendly digital products. My design philosophy focuses on clarity, simplicity, and usability.

</p>

</section>

<section class="services" id="services">

<h2>Services</h2>

<div class="services-container">

<div class="service-box">

<h3>Web UI Design</h3>

<p>Clean, modern interfaces designed for usability and engagement.</p>

</div>

<div class="service-box">

<h3>Mobile App Design</h3>

<p>Responsive and intuitive app designs for iOS and Android.</p>

</div>

<div class="service-box">

<h3>Prototyping</h3>

<p>Interactive wireframes and prototypes for seamless collaboration.</p>

</div>

</div>

</section>

<section class="portfolio" id="portfolio">

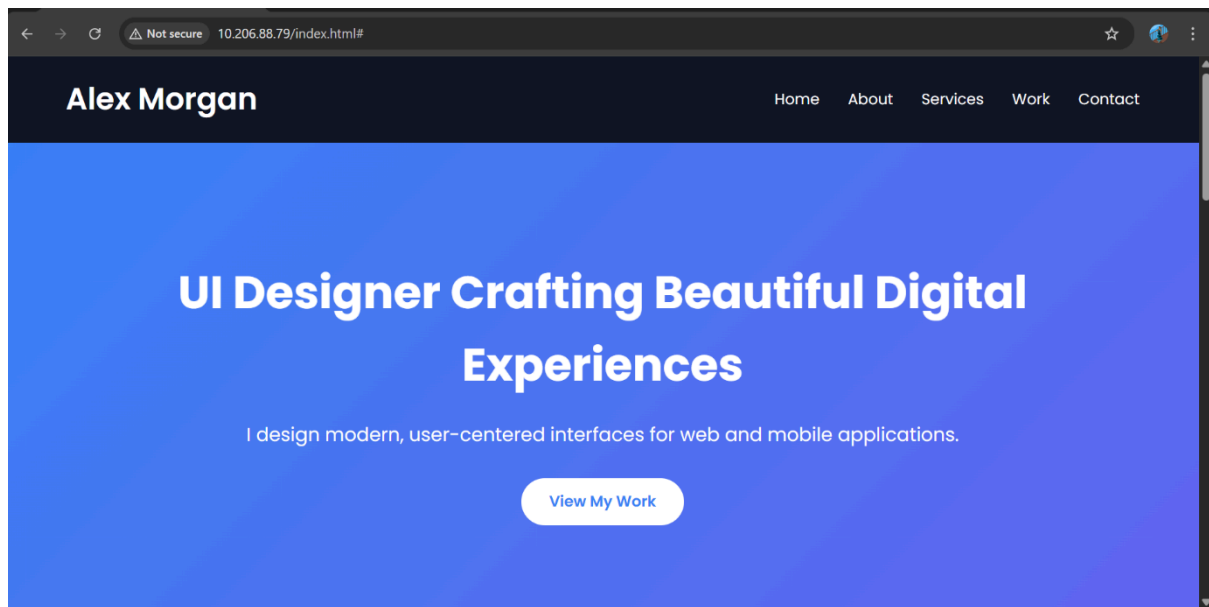
<h2>Portfolio</h2>

</section>

</body>

</html>

## Output



## Result

The web server was successfully configured on the Raspberry Pi. The web page was accessible through a browser using the Raspberry Pi's IP address. This project demonstrates that Raspberry Pi can function as a compact, cost-effective web server for embedded and IoT applications.